



BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Lymphatic System Note Sheets.

Muscle · Muscles of the Head, Neck, Trunk, and Arm · Heart · Blood Vessels · Blood

PART A

The brain dump prompts

All 7 lymphatic competencies with two drawing prompts each. Read this part once, then pick.

PART B

The note sheets

One sheet per competency. The competency and its two prompts on the left, one big drawing box on the right. Seven sheets.

How this packet works

Every lymphatic competency is on one sheet with two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice, and it is a good place to go the week before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before any video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is the point. It is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing. Some want a table. Some want two views side by side. The division is yours to make.
4. Words are allowed inside the drawing, as labels and inside little boxes with arrows between them. A sentence running the width of the box is not. If a box reads like a paragraph, redo it as a picture.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a table or a sequence, so a student who thinks in pictures and a student who thinks in categories both get a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set the same thing up on a tablet, and copy the prompt letter you chose at the top. A hand ruled sheet counts exactly the same as a printed one.

Lecture and lab

The lymphatic system is lecture only this term, so none of these sheets are on the lab practical. Check your drawings against the lymphatic notes and the video chapters.

PART A

The brain dump prompts.

Seven competencies from the lymphatic concept video. Each one gives you two prompts. Build one of them in that competency's box on your note sheet in Part B.

The lymphatic system

1 The lymphatic system and the pathway of lymph

MODULE 4

LECTURE

You should be able to: Define lymph, name the three jobs of the lymphatic system, and trace lymph one way from the tissue spaces back to the blood.

A FLOW MAP

Draw a blood capillary bed with tissue cells around it. Show fluid leaking out of the blood capillary into the tissue spaces and a lymphatic capillary picking it up. Then draw a chain of boxes with arrows: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein.

Then: Mark on your drawing where each of the three jobs happens: returning fluid, absorbing fat, and defense.

B DRAINAGE BODY MAP

Draw a body outline. Shade the area drained by the right lymphatic duct in one color and the area drained by the thoracic duct in another. Draw the cisterna chyli, the thoracic duct, and both ducts emptying where the subclavian and internal jugular veins meet.

Then: Write beside your right arm and your right leg the duct that drains each one.

2 Lymphatic capillaries and lacteals

MODULE 4

LECTURE

You should be able to: Compare lymphatic capillaries with blood capillaries, name where lymphatic capillaries are not found, and describe how lacteals absorb dietary fat.

A SIDE BY SIDE CAPILLARIES

Draw a blood capillary next to a lymphatic capillary. On the lymphatic one, show the blind end, the overlapping endothelial cells acting as flap valves, and the anchoring filaments. Draw what can get in: fluid, proteins, cells, and pathogens.

Then: Write three places lymphatic capillaries are not found, such as the central nervous system, beside your drawing.

B VILLUS CLOSE-UP

Draw one intestinal villus with a blood capillary network and a lacteal in its center. Use arrows to show sugars and amino acids going into the blood and fats going into the lacteal. Label the milky lymph as chyle.

Then: Add arrows that follow chyle from the lacteal to the cisterna chyli and up the thoracic duct.

3 Collecting vessels, trunks, and ducts

MODULE 4

LECTURE

You should be able to: Describe lymphatic collecting vessels, name the nine lymphatic trunks and the duct each one empties into, and explain how lymph moves without a pump.

A TRUNK MAP

Draw an outline of the trunk with the thoracic duct and the right lymphatic duct. Draw and label the nine trunks: the left and right lumbar, the intestinal, and the left and right bronchomediastinal, subclavian, and jugular trunks. Connect each one to the duct it empties into.

Then: Color the three trunks that empty into the right lymphatic duct.

B DRAINAGE TABLE

Draw a table with the nine trunks down the side. Across the top: the region it drains and the duct it empties into. Use a word or a tiny sketch in each cell, not a sentence.

Then: Below the table, draw a collecting vessel cut open to show its valves, and label three things that push lymph along.

4 Lymph node structure

MODULE 4

LECTURE

You should be able to: Describe the parts of a lymph node and the path lymph takes through it, including the capsule, cortex, follicles, medulla, sinuses, hilum, and afferent and efferent vessels.

A LABELED NODE

Draw a bean-shaped lymph node. Label the capsule, trabeculae, subcapsular sinus, cortex with follicles and germinal centers, medulla with medullary cords and sinuses, hilum, afferent vessels, and efferent vessels. Add arrows for the path of lymph.

Then: Count your afferent and efferent vessels and write beside the hilum why lymph slows down inside the node.

B CLUSTER MAP

Draw a body outline and mark the main node clusters: cervical, axillary, and inguinal, plus nodes in the thorax and abdomen. Draw one node zoomed in with a macrophage engulfing a bacterium and a lymphocyte beside it.

Then: Next to each cluster, write the body region it filters.

5 Lymphocytes, lymphatic tissue, and the primary organs

MODULE 4

LECTURE

You should be able to: Name the three types of lymphocytes and what each does, trace lymphopoiesis from the red bone marrow and thymus, and describe the structure of the thymus.

A LYMPHOCYTE FAMILY TREE

Start with one stem cell in the red bone marrow. Branch it into B cells that mature in the marrow, T cells that travel to the thymus to mature, and NK cells. Under each, write its share of circulating lymphocytes, its type of immunity, and what it attacks.

Then: Draw arrows from each mature cell to the secondary organs where it settles.

B THYMUS CLOSE-UP

Draw the thymus in the mediastinum in a child and in an adult. Then draw one thymic lobule with the cortex and medulla, thymic epithelial cells, and a thymic corpuscle.

Then: Label where T cells are still immature and where they are mature.

6 The spleen, tonsils, and MALT

MODULE 4

LECTURE

You should be able to: Describe the spleen's red pulp and white pulp, trace blood through the spleen, and locate the tonsils and other mucosa-associated lymphatic tissue.

A SPLEEN BLOOD FLOW

Draw the spleen in the upper left abdomen with the splenic artery and vein at the hilum. Then draw a close-up and trace blood with arrows: splenic artery, trabecular artery, central artery in white pulp, red pulp, venous sinuses, splenic vein.

Then: Label what the white pulp does and what the red pulp does, and write which organs take over after a splenectomy.

B GUARD POSTS MAP

Draw a side view of the head and neck with the pharyngeal, palatine, and lingual tonsils, then a loop of small intestine with Peyer patches and the appendix. Draw one lymphatic follicle with its germinal center.

Then: Draw a two-column box sorting every lymphatic organ into primary and secondary.

7 Lymphatic disorders

MODULE 4

LECTURE

You should be able to: Name edema, lymphedema, elephantiasis, lymphadenopathy, lymphoma, splenomegaly, ruptured spleen, and tonsillitis and identify the structure each one affects.

A DISORDER MAP

Draw a body outline. Place each disorder at the structure it affects, with a tiny sketch of what changes: edema, lymphedema, elephantiasis, lymphadenopathy, lymphoma, splenomegaly, ruptured spleen, and tonsillitis.

Then: Circle the disorders caused by blocked lymphatic drainage.

B DISORDER TABLE

Draw a table with the eight disorders down the side. Across the top: the structure affected, what changes, and one cause. Use a word or a tiny sketch in each cell.

Then: Star the disorder that makes contact sports dangerous during mononucleosis and say which organ is at risk.

PART B

The note sheets.

One sheet per competency, 7 sheets. The competency and both prompts sit on the left so nothing else has to be printed beside it. The box on the right is the whole assignment.

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1

COMP

MODULE 4 · LYMPHATIC SYSTEM

The lymphatic system and the pathway of lymph

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PICK ONE AND DO THAT ONE

☐ A FLOW MAP

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Then draw a chain of boxes with arrows: lymphatic capillary, collecting vessel, lymph node, trunk, duct, subclavian vein.

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Then: Write beside your right arm and your right leg the duct that drains each one.

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Dr. Sharilyn Rennie

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